Swati Aggrawal

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Hiring Team, ShopFully
Machine Learning Engineer

Dear Hiring Team,

I am writing to apply for the Machine Learning Engineer role at ShopFully. The combination of scale ,100 million daily events, 200 million shoppers and the technical depth required, spanning multimodal models, semantic similarity, and production MLOps, is exactly the kind of environment where I want to grow. I am an IEEE-published ML researcher and engineer currently completing my MSc in Computer Science at Sapienza Università di Roma (May 2026), and I believe my profile maps closely to what you are looking for.

On the engineering side, I built and deployed **DocDelta**, a production semantic document comparison API, end to end and solo: FastAPI, Dockerised NLP pipeline, deployed on AWS ECS Fargate, and monitored via CloudWatch. I profiled real latency benchmarks (~840ms warm on CPU-only PyTorch) and identified concrete optimisation paths. This experience gave me first-hand familiarity with the full MLOps lifecycle you describe — from experiment to production deployment serving live inference.

On the research side, I built a **Lossy Text Compressor** using GTR-T5 embeddings and vec2text inversion, benchmarking 8 quantization levels and achieving cosine similarity 0.964 and ROUGE-L 0.776 at optimal 6-bit compression. My MSc thesis investigates pretrained Vision Transformers under continual learning directly relevant to your work on multimodal models and content understanding from offer images and unstructured text. I also published a peer-reviewed IEEE survey paper on ML for air quality prediction, cited 6 times since 2023.

What draws me specifically to ShopFully is the **"Learn Loudly"** and **"Always Build Better"** values, they reflect how I already work. The challenge of building AI systems that process 100 million daily events, extract structured data from unstructured retail content, and power personalised recommendations at scale is genuinely exciting to me. I am comfortable in Python, PyTorch, FastAPI and HuggingFace, and eager to grow into the multimodal and recommendation system areas your team works on.

I am based in Rome and fully open to the hybrid or remote model across Italy. I would love the opportunity to discuss how I can contribute to ShopFully's mission.

Thank you for your time and consideration.

Best regards,
Swati Aggrawal
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